PROGRAMMING FUNDAMENTALS (CS-1002)

ASSIGNMENT 1

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Section: BAI-1B

PART A:

Question 1:

ALGORITHM:

1. START
2. Take input from user “n”
3. Loop While n is greater than or equal to 1
    1. Output n
    2. decrement n by 1, n = n-1
4. End Loop
5. END

PSEUDOCODE:

```
Step 1.     START
Step 2.     INPUT INTEGER n from user
Step 3.     WHILE LOOP, n >= 1
Step 4.         PRINT n
Step 5.         n = n – 1
Step 6.     END LOOP
Step 7.     END
```

Question 2:

ALGORITHM:

1. START
2. Take input from user “year”
3. If year / 4 reminder is 0 and year / 100 reminder is non-zero
    1. Then Output “Leap Year”
4. Else If year / 400 reminder is 0,
    1. Then Output “Leap Year”
5. Else Output “Not a Leap Year”
6. END

PSEUDOCODE:

```
Step 1.     START
Step 2.     INPUT INTEGER year from user
Step 3.     IF year % 4 == 0 AND year % 100 != 0,
Step 4.         THEN PRINT “Leap Year”
Step 5.     ELSE IF year % 400 == 0
Step 6.         THEN PRINT “Leap Year”
Step 7.     ELSE PRINT “Not a Leap Year”
Step 8.     END
```

Question 3:

ALGORITHM:

1. START
2. Take input number from user, num
3. If num / 10 gives 0 Then output “single digit”
4. Else If num / 100 gives 0 Then output “Double Digit”
5. Else If num / 1000 gives 0 Then output “Triple Digit”
6. Else Output “more than 3 digits”
7. END

PSEUDOCODE:

```
Step 1.     START
Step 2.     INPUT INTEGER num from user
Step 3.     IF num / 10 == 0
Step 4.         THEN PRINT “Single Digit”
Step 5.     ELSE IF num / 100 == 0
Step 6.         THEN PRINT “Double Digit”
Step 7.     ELSE IF num / 1000 == 0
Step 8.         THEN PRINT “Triple Digit”
Step 9.     ELSE PRINT “More than 3 Digits”
Step 10.    END
```

Question 4:

ALGORITHM:

1. START
2. Set n = 1
3. Loop While n is less than or equal to 10
    1. Set i = 1
    2. If n / 3 gives remainder non-zero
        1. Then Loop While i <= 10
            1. Set p = n*i
            2. Output n x i = p
            3. i = i + 1
        2. End Loop
        3. n = n + 1
    3. Else n = n + 1
4. End Loop
5. END

PSEUDOCODE:

```
Step 1.     START
Step 2.     SET n = 1
Step 3.     WHILE LOOP, n <= 10
Step 4.       SET i = 1
Step 5.       IF n % 3 != 0
Step 6.         WHILE LOOP, i <= 10
Step 7.           SET INTEGER p = n * i
Step 8.           PRINT n x i = p
Step 9.           i = i + 1
Step 10.        END LOOP
Step 11.        n = n + 1
Step 12.      ELSE n = n + 1
Step 13.    END LOOP
Step 14.    END
```

Question 5:

ALGORITHM:

1. START
2. Take input from user, num
3. Set r = 0, sum = 0
4. Loop While (num / 10) is not 0
5. sum = 0
    1. Loop while num is NOT 0
        1. r = remainder of num / 10
        2. num = num / 10
        3. sum = sum + r
    2. End Loop
    3. Output sum
    4. num = sum
6. End Loop
7. END

PSEUDOCODE:

```
Step 1.     START
Step 2.     INPUT INTEGER num from user
Step 3.     SET r = 0, sum = 0
Step 4.     WHILE LOOP, num / 10 != 0
Step 5.       SET sum = 0
Step 6.       WHILE LOOP, num != 0
Step 7.         r = num % 10
Step 8.         num = num / 10
Step 9.         sum = sum + r
Step 10.      END LOOP
Step 11.      PRINT sum
Step 12.       num = sum
Step 13.    END LOOP
Step 14.    END
```

PART B:

Question 1:

ALGORITHM:

1. START
2. Set total revenue = 0,
3. Input number of Guests (n)
4. Loop n times
    1. Input nights staying
    2. Input room type (1. Standard, 2. Deluxe, 3. Suite)
    3. Input Season (1. Peak, 2. Off Peak)
    4. Check season
        1. for Peak Season(1), set standard rate = 5000, deluxe rate = 8000, suite rate = 12000
        2. for Off Peak Season(2), set standard rate = 3000, deluxe rate = 5000, suite rate = 8000
    5. Check room type
        1. for Standard (1), set rate = standard rate
        2. for Deluxe (2), set rate = deluxe rate
        3. for Suite (3), set rate = suite rate
    6. if nights staying is more than 7
        1. then calculate discount = (rate*nights)*0.15
    7. else set discount = 0
    8. calculate final price = (rate * nights) – discount
    9. Display final price
    10. total revenue = total revenue + final price
5. End loop
6. Display total revenue
7. END

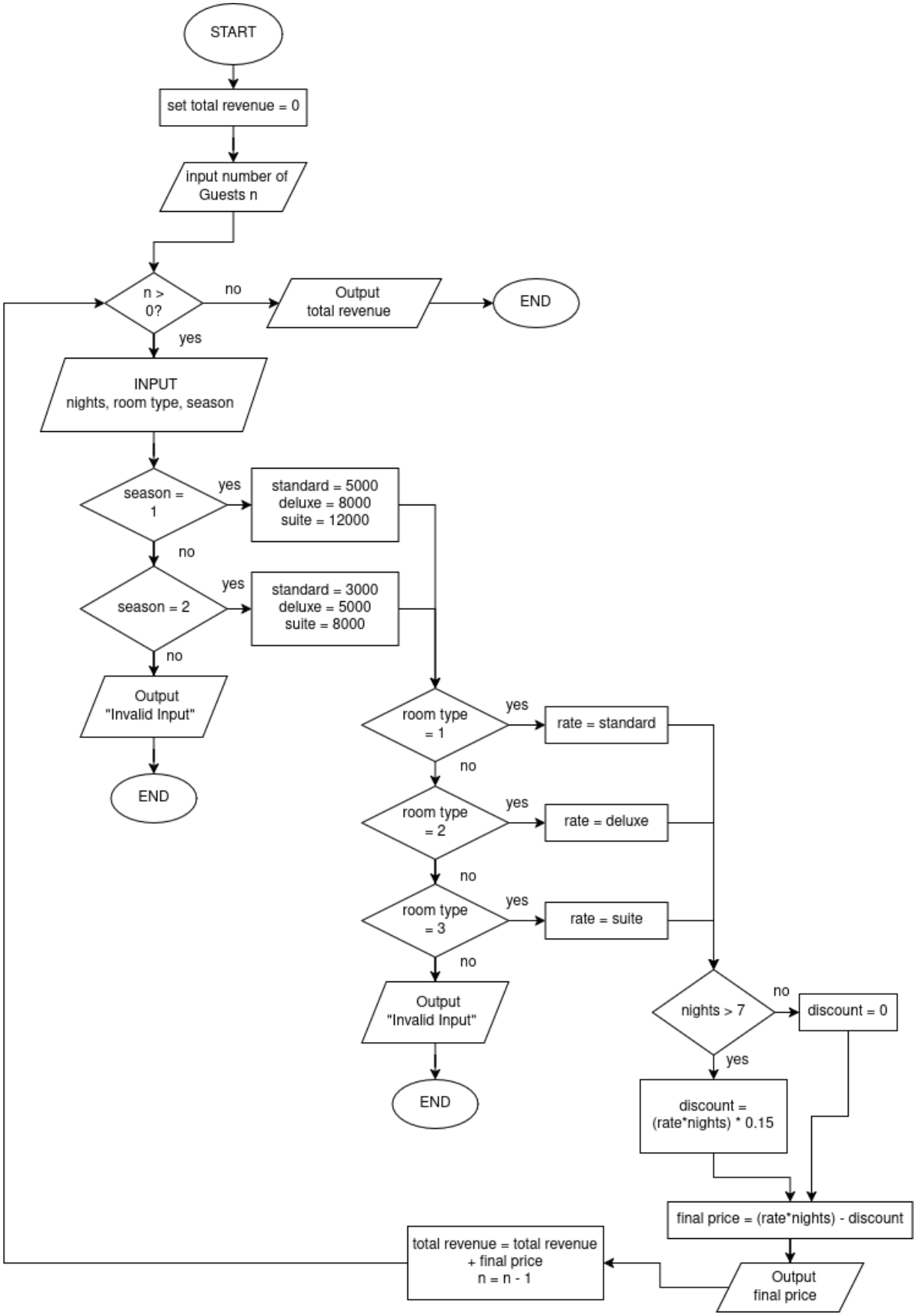
PAC:

Given Inputs	Required Results
Number of Guests Season (Peak/Off Peak) Nights Staying Room Type (Standard/Deluxe/Suite)	Final Price Total Revenue
Processing Required	**Alternate Solutions**
1. initialize variables, set total revenue = 0 2. Loop for each guest (n) 3. Input Season, Nights, room type 4. identify rates using if-else statements 5. apply discount if nights more than 7, otherwise set discount to 0 6. calculate and Display final price = (rate × nights) − discount 7. Total revenue = Total revenue + final price *(looped for N number of Guests)* *8. display total revenue after Loop Ends*	1. Use *Switch-Case* method for Season Identification and their respective calculation 2. Calculate Final price with [***(rate*nights)*0.85***] for 7 or more nights

IPO CHART:

Input	Processing	Module Reference	Output
Number of Guests Season *(Peak/Off Peak)* Nights Staying Room Type *(Standard/Deluxe/Suite)*	1. Initialize total revenue = 0 2. Input number of guests (n) 3. Loop for each guest (n times): 4. Input season, nights staying and room type 5. Identify rates using the season 6. Select the rate using the room type 7. If nights > 7, 8. discount = (rate × nights) × 0.15, 9. otherwise discount = 0 10. Final price = (rate × nights) − discount 11. Display final price 12. Total revenue = total revenue + final price 13. After the loop ends, display total revenue	SET READ WHILE LOOP READ SWITCH-CASE SWITCH-CASE IF CALC ELSE CALC PRINT CALC PRINT	Final Price Total Revenue

FLOWCHART:


START
set total revenue = 0
input number of Guests n
n > 0?
no
Output total revenue
END
yes
INPUT nights, room type, season
season = 1
yes
standard = 5000
deluxe = 8000
suite = 12000
no
season = 2
yes
standard = 3000
deluxe = 5000
suite = 8000
no
Output "Invalid Input"
END
room type = 1
yes
rate = standard
no
room type = 2
yes
rate = deluxe
no
room type = 3
yes
rate = suite
no
Output "Invalid Input"
END
nights > 7
no
discount = 0
yes
discount = (rate*nights) * 0.15
final price = (rate*nights) - discount
Output final price
total revenue = total revenue + final price
n = n - 1

PSEUDOCODE:

```
Step 1.   START
Step 2.   SET total_revenue = 0
Step 3.   INPUT INTEGER n from user
Step 4.   WHILE LOOP, n > 0
Step 5.     INPUT INTEGER nights from user
Step 6.     INPUT INTEGER room_type from user
Step 7.     INPUT INTEGER season from user
Step 8.     SWITCH season
Step 9.       CASE 1: SET standard = 5000, deluxe = 8000, suite = 12000
Step 10.      CASE 2: SET standard = 3000, deluxe = 5000, suite = 8000
Step 11.    END SWITCH
Step 12.    SWITCH room_type
Step 13.      CASE 1: SET rate = standard
Step 14.      CASE 2: SET rate = deluxe
Step 15.      CASE 3: SET rate = suite
Step 16.    END SWITCH
Step 17.    IF nights > 7
Step 18.      SET discount = (rate * nights) * 0.15
Step 19.    ELSE
Step 20.      SET discount = 0
Step 21.    END IF
Step 22.    SET final_price = (rate * nights) - discount
Step 23.    PRINT "Final price =", final_price
Step 24.    total_revenue = total_revenue + final_price
Step 25.    n = n - 1
Step 26.  END LOOP
Step 27.  PRINT "Total revenue =", total_revenue
Step 28.  END
```

C CODE:

```c
#include <stdio.h>

int main(){

    int n, nights, room_type, season;
    int standard, deluxe, suite, rate;
    float discount, final_price;
    float total_revenue = 0;
    int i = 1;

    printf("Enter number of guests: ");
    scanf("%d", &n);

    while (n > 0){
        printf("-----GUEST %d-----\n", i);
        printf("Enter nights staying: ");
        scanf("%d", &nights);
        printf("1. Standard\n2. Deluxe\n3. Suite\n");
        printf("Enter the Room Type: ");
        scanf("%d", &room_type);
```

```
        printf("Enter season (1. Peak, 2. Off Peak): ");
        scanf("%d", &season);

        switch (season){
            case 1:
                standard = 5000; deluxe = 8000; suite = 12000;
                break;
            case 2:
                standard = 3000; deluxe = 5000; suite = 8000;
                break;
            default:
                printf("Invalid Input!");
                break;
        }

        switch (room_type){
            case 1:
                rate = standard;
                break;
            case 2:
                rate = deluxe;
                break;
            case 3:
                rate = suite;
                break;
            default:
                printf("Invalid Input!");
                break;
        }

        if (nights > 7){
            discount = (rate * nights) * 0.15;
        }
        else{
            discount = 0;
        }

        final_price = (rate * nights) - discount;
        printf("Final price = %.2f\n", final_price);
        total_revenue = total_revenue + final_price;

        n = n - 1;
        i = i + 1;
    }

    printf("Total revenue = %.2f\n", total_revenue);
    return 0;
}
--------------------------------------------------------------------------------
```

Question 2:

ALGORITHM:

1. START
2. Set current floor = 0
3. Input number of requests (n)
4. Loop While n > 0
5. Input floor request
6. If floor request is greater than current floor
7. Then Output "Moving Up"
8. Else If floor request is less than current floor
9. Then Output "Moving Down"
10. Else Output "Doors Opening"
11. current floor = floor request
12. n = n – 1
13. End Loop
14. END

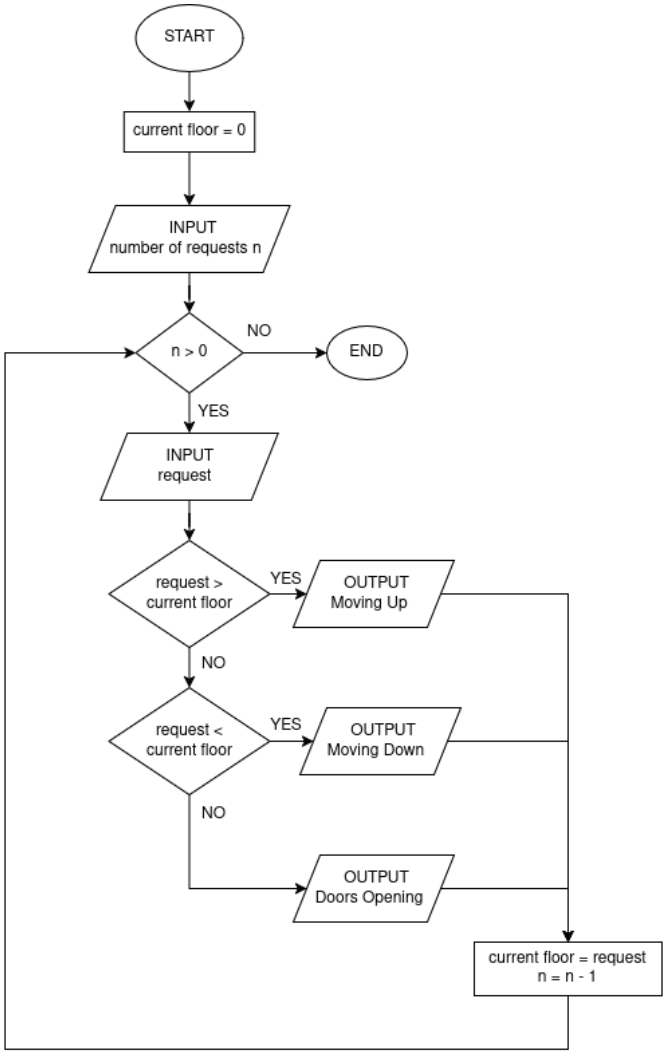
PAC:

Given Inputs	Required Results
Floor request	Direction of elevator
Processing Required	**Alternate Solutions**
1. Initialize floor 0 2. input request amount n 3. loop for n number of floor requests 4. input floor request 5. if floor request is greater than floor, say moving up and floor = floor request 6. else if floor request is less than floor, say moving down and floor = floor request 7. else floor request is same as floor, say doors opening 8. n = n - 1 9. end loop	1. Store requests in an Array and process one by one 2. use a function move(floor, request) that changes the floor

IPO CHART:

Input	Processing	Module Reference	Output
Number of requests n Floor Request(in each loop)	1. Initialize floor = 0 2. input number of requests n 3. loop for n number of requests 4. input floor request 5. if request > floor 6. then Display Moving Up 7. floor = request 8. else if request < floor 9. then Display Moving Down 10. floor = request 11. else floor = request 12. Display Doors Opening 13. n = n - 1 14. End Loop	ASSIGN READ WHILE LOOP READ IF PRINT ASSIGN ELSE IF PRINT ASSIGN ELSE PRINT CALC END LOOP	Direction of Movement

FLOWCHART:



PSEUDOCODE:

Step 1. START
Step 2. SET current_floor = 0
Step 3. INPUT INTEGER n from user
Step 4. WHILE LOOP, n > 0
Step 5. INPUT INTEGER request from user
Step 6. IF request > current_floor
Step 7. PRINT "Moving Up"
Step 8. ELSE IF request < current_floor
Step 9. PRINT "Moving Down"
Step 10. ELSE
Step 11. PRINT "Doors Opening"
Step 12. END IF
Step 13. SET current_floor = request
Step 14. n = n - 1
Step 15. END LOOP
Step 16. END

C CODE

```
#include <stdio.h>

int main(){

    int n, request;
    int current_floor = 0;

    printf("Enter number of requests: ");
    scanf("%d", &n);

while (n > 0){
        printf("Enter floor request: ");
        scanf("%d", &request);

        if (request > current_floor){
            printf("Moving Up\n");
        }

        else if (request < current_floor){
            printf("Moving Down\n");
        }

        else{
            printf("Doors Opening\n");
        }

        current_floor = request;
        n = n - 1;
    }

    return 0;
}
```

Question 3:

ALGORITHM

1. START
2. Input number of students (n)
3. Loop While n > 0
   1. Set sum = 0, failed_subject = 0, count = 1
   2. Loop While count <= 5
      1. Input mark
      2. sum = sum + mark
      3. If mark < 33
         1. Then failed_subject = 1
      4. count = count + 1
   3. End Loop
   4. average = sum / 5
   5. Output average
   6. If failed_subject is 1
      1. Then Output "Fail - Subject Deficiency"
   7. Else If average >= 80
      1. Then Output "Distinction"
   8. Else If average >= 60
      1. Then Output "Pass"
   9. Else Output "Fail"
   10. n = n - 1
4. End Loop
5. END

PAC:

Given Inputs	Required Results
number of students (N) 5 marks per student	average and result for each student
Processing Required	**Alternate Solutions**
1. Input number of students 2. loop for n students 3. set sum = 0, failed subject = 0, count = 1 4. loop for 5 marks 5. input marks 6. sum = sum + marks 7. if marks < 33, failed subject = 1 8. count = count + 1 9. average = sum / 5 10. display average 11. if failed subject = 1, Display fail – SubjectDefeiciency 12. Else if average >= 80, display distinction 13. else if average >= 60, display Pass 14. Else display fail 15. n = n - 1	

IPO CHART:

Input	Processing	Module Reference	Output
unmber of students (N) 5x marks per student	1. Input number of students 2. loop for n students 3. set sum = 0, failed subject = 0, count = 1 4. loop for 5 marks 5. input marks 6. sum = sum + marks 7. if marks < 33, failed subject = 1 8. count = count + 1 9. average = sum / 5 10. display average 11. if failed subject = 1, Display fail – SubjectDefeiciency 12. Else if average >= 80, display distinction 13. else if average >= 60, display Pass 14. Else display fail 15. n = n - 1	READ WHILE LOOP SET WHILE LOOP READ CALC IF CALC CALC PRINT IF ELSE IF ELSE IF ELSE CALC	average and result for each student

FLOWCHART:

PSEUDOCODE:

```
Step 1.     START
Step 2.     INPUT INTEGER n from user
Step 3.     WHILE LOOP, n > 0
Step 4.       SET sum = 0, failed_subject = 0, count = 1
Step 5.       WHILE LOOP, count <= 5
Step 6.         INPUT INTEGER mark from user
Step 7.         sum = sum + mark
Step 8.         IF mark < 33
Step 9.           SET failed_subject = 1
Step 10.        END IF
Step 11.        count = count + 1
Step 12.      END LOOP
Step 13.      SET average = sum / 5
Step 14.      PRINT "Average =", average
Step 15.      IF failed_subject == 1
```

```
Step 16.            PRINT "Fail - Subject Deficiency"
Step 17.         ELSE IF average >= 80
Step 18.            PRINT "Distinction"
Step 19.         ELSE IF average >= 60
Step 20.            PRINT "Pass"
Step 21.         ELSE
Step 22.            PRINT "Fail"
Step 23.         END IF
Step 24.         n = n - 1
Step 25.      END LOOP
Step 26.      END
```

C CODE:

```
#include <stdio.h>

int main(){
   int n, mark, sum, failed_subject, count;
   float average;

   printf("Enter number of students: ");
   scanf("%d", &n);

   while (n > 0){
      sum = 0;
      failed_subject = 0;
      count = 1;

      while (count <= 5){
         printf("Enter mark %d: ", count);
         scanf("%d", &mark);
         sum = sum + mark;

         if (mark < 33)
         {
            failed_subject = 1;
         }

         count = count + 1;
      }

      average = sum / 5.0;
      printf("Average = %.2f\n", average);

      if (failed_subject == 1){
         printf("Fail - Subject Deficiency\n");
      }

      else if (average >= 80){
         printf("Distinction\n");
      }

      else if (average >= 60){
         printf("Pass\n");
      }

      else{
         printf("Fail\n");
```

```
        }

        n = n - 1;
    }

    return 0;
}
```

Question 4:

ALGORITHM

1. START
2. Input quantity (q), price per item (p), discount percentage (d), tax percentage (t)
3. If q <= 0
4. Then Output "Invalid quantity: must be greater than 0" and terminate
5. Else If p <= 0
6. Then Output "Invalid price: must be greater than 0" and terminate
7. Else If d < 0 or d >= 100
8. Then Output "Invalid discount: must be from 0 to below 100" and terminate
9. Else If t < 0 or t >= 100
10. Then Output "Invalid tax: must be from 0 to below 100" and terminate
11. Else
12. Calculate subtotal: s = q * p
13. Calculate discounted amount: a = s - (s * d) / 100
14. Calculate final bill = a + (a * t) / 100
15. Display the bill document: subtotal, discount, discounted amount, tax, final bill
16. End If
17. END

PAC:

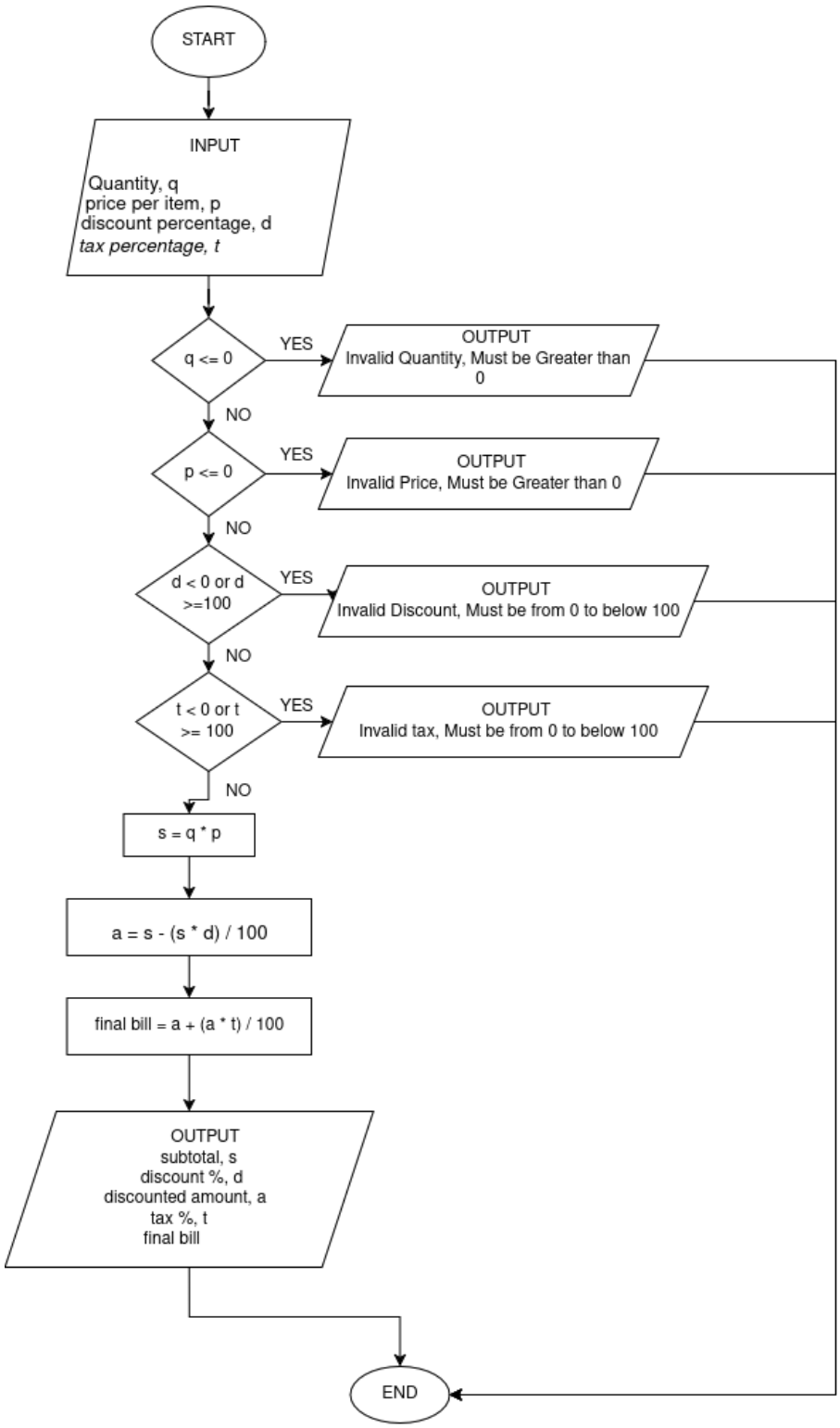
Given Inputs	Required Results
Quantity, q price per item, p discount percentage, d tax percentage, t	Subtotal, s Discounted Amount, a Final Bill
Processing Required	**Alternate Solutions**
1. Input quantity, price per item, discount %age and tax %age 2. Validate the inputs 3. If any value is invalid, display an error message and terminate 4. Otherwise calculate the subtotal: s = q × p 5. Calculate the discounted amount: a = s − (s × d) / 100 6. Calculate the final bill: final bill = a + (a × t) / 100 7. display the bill document (subtotal, discount, discounted amount, tax, final bill)	1. Write each formula as a separate function: subtotal(q, p), discounted(s, d) and final_bill(a, t) 2. Calculate the final bill in one combined statement

IPO CHART:

Input	Processing	Module Reference	Output
Quantity, q price per item, p discount percentage, d *tax percentage, t*	1. Input quantity, price per item, discount %age and tax %age 2. Validate the inputs 3. If any value is invalid, display an error message and terminate 4. Else calculate the subtotal: s = q×p 5. Calculate the discounted amount: a = s − (s × d) / 100 6. Calculate the final bill: final bill =	READ IF PRINT ELSE, CALC CALC CALC	Subtotal, s Discounted Amount, a Final Bill

	a + (a × t) / 100 7. display the bill document (subtotal, discount, discounted amount, tax, final bill)	PRINT	

FLOWCHART:



PSEUDOCODE:

```
Step 1.   START
Step 2.   INPUT INTEGER q from user
Step 3.   INPUT FLOAT p from user
Step 4.   INPUT FLOAT d from user
Step 5.   INPUT FLOAT t from user
Step 6.   IF q <= 0
Step 7.      PRINT "Invalid quantity: must be greater than 0"
Step 8.   ELSE IF p <= 0
Step 9.      PRINT "Invalid price: must be greater than 0"
Step 10.  ELSE IF d < 0 OR d >= 100
Step 11.     PRINT "Invalid discount: must be from 0 to below 100"
Step 12.  ELSE IF t < 0 OR t >= 100
Step 13.     PRINT "Invalid tax: must be from 0 to below 100" D
Step 14.  ELSE
Step 15.     SET s = q * p
Step 16.     SET a = s - (s * d) / 100
Step 17.     SET final_bill = a + (a * t) / 100
Step 18.     PRINT "Subtotal =", s
Step 19.     PRINT "Discount (%) =", d
Step 20.     PRINT "Discounted amount =", a
Step 21.     PRINT "Tax (%) =", t
Step 22.     PRINT "Final bill =", final_bill
Step 23.  END IF
Step 24.  END
```

C CODE:

```c
#include <stdio.h>

int main(){
    int q;
    float p, d, t;
    float s, a, final_bill;

    printf("Enter quantity: ");
    scanf("%d", &q);
    printf("Enter price per item: ");
    scanf("%f", &p);
    printf("Enter discount percentage: ");
    scanf("%f", &d);
    printf("Enter tax percentage: ");
    scanf("%f", &t);

    if (q <= 0){
        printf("Invalid quantity: must be greater than 0\n");
    }

    else if (p <= 0){
        printf("Invalid price: must be greater than 0\n");
    }

    else if (d < 0 || d >= 100){
        printf("Invalid discount: must be from 0 to below 100\n");
```

```
    }

    else if (t < 0 || t >= 100){
        printf("Invalid tax: must be from 0 to below 100\n");
    }

    else{
        s = q * p;
        a = s - (s * d) / 100;
        final_bill = a + (a * t) / 100;

        printf("\n----- BILL -----\n");
        printf("Subtotal = %.2f\n", s);
        printf("Discount (%%) = %.2f\n", d);
        printf("Discounted amount = %.2f\n", a);
        printf("Tax (%%) = %.2f\n", t);
        printf("Final bill = %.2f\n", final_bill);
    }

    return 0;
}
```